

Integrals in Context: Average Value, Accumulation, and Motion

Average value, accumulated change from a rate, and motion along a line

Directions

- The first three problems are a warm-up, one on each idea. After that the three ideas are mixed — decide which one a problem is asking for before you write anything.
- Three questions to tell them apart: *How much accumulated?* integrate the rate. *What was it on average?* integrate, then divide by the length of the interval. *How far did it actually travel?* integrate the absolute value of the velocity.
- Where an answer carries a unit, the answer line names it.

1. Find the average value of $f(x) = 3x^2$ on the interval $[0, 4]$.

Answer: _____

2. Water flows into a tank at the rate $r(t) = 4t + 2$ litres per minute, where t is measured in minutes. How much water enters the tank during the first 6 minutes?

Answer: _____ L

3. A particle moves along a straight line with velocity $v(t) = 3t^2 - 12$ metres per second. Find the particle's displacement between $t = 0$ and $t = 4$ seconds.

Answer: _____ m

4. Find the average value of $g(x) = \sin x$ on the interval $[0, \pi]$. Give an exact answer and a decimal to three places.

Answer: _____

5. Sand is poured onto a pile at the rate $r(t) = 9 - t$ kilograms per minute, for $0 \leq t \leq 9$ minutes. How much sand is added to the pile over those 9 minutes?

Answer: _____ kg

6. A particle moves along a straight line with velocity $v(t) = t^2 - 4t + 3$ metres per second, for $0 \leq t \leq 4$ seconds.

(a) Find the times in $[0, 4]$ at which the particle changes direction.

Answer: _____

(b) Find the total distance the particle travels over $[0, 4]$.

Answer: _____ m

7. The temperature inside a greenhouse is modelled by $T(t) = 18 + 6 \sin\left(\frac{\pi t}{12}\right)$ degrees Celsius, where t is the number of hours after midnight. Find the average temperature over the first 12 hours. Give an exact answer and a decimal to three places.

Answer: _____

8. Water leaks from a storage tank at the rate $r(t) = 20e^{-0.1t}$ litres per hour, where t is in hours. How much water leaks out during the first 10 hours? Give an exact answer and a decimal to two places.

Answer: _____ L

9. A particle moves along the x -axis with velocity $v(t) = 4 - t$ metres per second, and its position at time $t = 0$ is $x = 3$ metres. Time is measured in seconds.

(a) At what time does the particle change direction?

Answer: _____

(b) Find the particle's position at $t = 6$.

Answer: _____ m

10. A particle moves along the x -axis with velocity $v(t) = t^2 - 6t + 8$ metres per second for $0 \leq t \leq 5$ seconds, starting at $x = 1$ metre.

- (a) Find the total distance the particle travels on $[0, 5]$.

Answer: _____ m

- (b) Find the particle's average velocity on $[0, 5]$.

Answer: _____ m/s

- (c) Explain why the average velocity is not the same as the average of the largest and smallest velocities on the interval.